SEPARATED AND UNIVERSALLY CLOSED QUASI-COMPACT QUASI-SEPARATED MORPHISMS CHARACTERIZED BY THE VALUATIVE CRITERION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the author's typo 'valutative' in one sentence; the text is delivered verbatim rather than corrected, since no rewriting of author text is permitted.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Uniqueness part of the criterion implies separatedness for quasi-separated morphisms. Lemma lemma-valuative-criterion-separatedness, used at proof lines 9270-9273. Gives one of the two linking implications.
- Separated morphisms satisfy the uniqueness part of the criterion. Lemma lemma-separated-implies-valuative, used at proof lines 9274-9276. Gives the other linking implication.
- Existence part of the criterion for quasi-compact quasi-separated morphisms. Lemma lemma-characterize-universally-closed-separated, used at proof lines 9277-9281. Is the input from which the three-way equivalence follows formally.
- The three formulations of the valuative criterion. Definition definition-valuative-criterion and Schemes, Definition schemes-definition-valuative-criterion, used at proof lines 9254-9266. Fix the conditions being compared.

Source reference. Original definition 03IX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4619–4619. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0A3Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12756–12756. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4690–4690. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

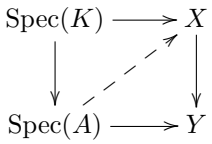
Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03KV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4691–4691. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 01KD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2046–2046. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is quasi-compact and quasi-separated. Then the following are equivalent*

(1) *f is separated and universally closed,*
(2) *the valuative criterion holds as in Definition 03IX,*
(3) *given any commutative solid diagram*



where A is a valuation ring with field of fractions K, there exists a unique dotted arrow, i.e., f satisfies the valuative criterion as in Schemes, Definition 01KD.

Proof. Since f is quasi-separated, the uniqueness part of the valutative criterion implies f is separated (Lemma 03KV). Conversely, if f is separated, then it satisfies the uniqueness part of the valuative criterion (Lemma 03KU). Having said this, we see that in each of the three cases the morphism f is separated and satisfies the uniqueness part of the valuative criterion. In this case the lemma is a formal consequence of Lemma 0A3Y. □